

Preliminary Insight Development: Used Car Prices in India

COMM5000 Data Literacy · Case Study Project · Milestone 1 · Term 3, 2026

Word count: 987 (headings included; tables, figures, captions, notes, references and appendix excluded)

1 Introduction

India's used car market now outsells the new car market, yet price appraisal still relies heavily on manual valuation (UNSW Business School 2026, p. 4). Our client, an online marketplace, needs a fast and consistent price estimate at appraisal to support instant cash offers, spot mispriced listings and forecast stock turnover. An offer set too high erodes margin and one set too low loses the seller, so the client needs a typical price and its likely spread.

This report explores the 1,005,000 listings in the course dataset (Stable Space 2026) against the two project objectives (UNSW Business School 2026, p. 4): how Car_Price relates to the other variables, including brand segment, and how the methods could be improved. Section 2 presents the analysis and Section 3 the findings and plan.

2 Data Summaries and Descriptive Statistics

2.1 Data preparation

The analysis uses the workbook as randomised on 26 September 2026 (Dataset sheet, cell B6), with every number calculated by Excel formulas. Labels were corrected only where the intended category was obvious, and no listing was deleted (Table 1). 'Unknown' entries were kept as a category because they affect 28.2% of rows and cannot be recovered. No cell is blank because the macro fills gaps (Stable Space 2026, cell B4) and sets negative draws to zero, so technical zeros were treated as missing.

The randomising macro (Franch 2026, slide 7) rewrites five numeric columns in its code, so repeated listings no longer look identical. Matching rows on the other 15 fields gives 31,575 identical pairs, about 4,750 more than the 26,821 expected by chance. These possible repeats cannot be told apart from chance matches, so they were flagged but kept. Keeping one row per match moves the median by only 0.01% (Table A4).

The guide treats more than 1,000,000 km as implausible (UNSW Business School 2026, p. 7). These 6,134 listings are close to the 6,102 expected from a log-normal fit, so they may be a genuine tail, but were left out of the usage analysis. A few isolated values, such as 4.6 hp, are extreme but too few to matter (Table 1).

Table 1 Data problems and how they were treated

Problem	Found with	Size	Treatment
Brand labels with extra spaces (e.g. ' BMW ')	TRIM, EXACT	10,040 rows (2,334 luxury)	Trimmed; 13 brands
Fuel_Type variants (PETROL, petrol, diesel with spaces, hybridd, elektrik)	LOWER, TRIM, lookup table	20,123 rows	Mapped to CNG, Diesel, Electric, Hybrid, Petrol; Unknown kept
Other text fields (Model, Transmission, Owner_Type, Color, City)	LEN, TRIM, category lists	No variants found	None needed
'Unknown' in Fuel_Type, Transmission, Color, City	COUNTIF	78,733; 80,385; 80,419; 80,430. 283,648 rows have at least one	Kept as a category
Blank cells	COUNTBLANK	0	None; the macro fills gaps
Zero technical values (the macro sets negative draws to 0)	COUNTIF	Mileage 4; Engine 12; Horsepower 10	Blank for that field only
Technical values beyond mean $\pm$ 3 SD	AVERAGEIF, STDEV.S, COUNTIFS	Mileage 36; Engine 52; Horsepower 33 (raw maxima 40.8 km/l, 5,502 cc, 393 hp)	Blank for that field only; far from the main range
Isolated values inside the $\pm$ 3 SD limits	MIN, COUNTIFS	hp: 44 < 50, 67 > 190; cc: 60 < 850, 97 > 2,250; km/l: 42 < 11, 37 > 25.5; price: 44 < INR 20,000	Kept; too few to move any median

Problem	Found with	Size	Treatment
Identical rows on all 20 fields	FREQUENCY on Car_Price	0	None needed
Repeated rows on the 15 non-randomised fields	Sorted key, run count, category shares	31,575 pairs in 29,640 repeat rows; 26,821 pairs expected by chance	Flagged and kept
Kms_Driven above 1,000,000 km	COUNTIF; NORM.DIST on LN(km)	6,134 rows, up to 5.8 million km (6,102 expected)	Excluded from usage analysis only
Extreme Car_Price (3 × IQR on log scale)	QUARTILE, COUNTIF	1 low, 2 high	Kept
Year and Registration_Age consistency	SUMPRODUCT	Sum is 2025 in all rows	Age used as supplied

Source: calculated from Stable Space (2026). A group of k identical rows gives $k(k - 1)/2$ pairs and $k - 1$ repeat rows. The chance count multiplies, over 13 fields, the probability that two random rows share a category (sum of squared shares); Brand and Year are left out because Model and Registration_Age already fix them. Limits for isolated values are the edges of the dense main range of each frequency distribution.

2.2 The variables

Table A1 describes all 20 fields. Car_Price is in Indian rupees (INR), and Year and Registration_Age carry the same information. Registration_Age is spread evenly over 1 to 25 years, and Engine_CC, Horsepower and Mileage_kmpl are roughly flat across their main range, so their modes in Table 2 are not typical values.

2.3 Distribution of Car_Price

Car_Price is strongly right-skewed (Table 2; skewness 4.9). The long right tail lifts the mean of INR 1,015,107 to 50.7% above the median of INR 673,522, and 67.5% of listings sit below the mean (Figure 1a). The median and interquartile range (INR 365,843 to 1,240,714) therefore describe a typical car better and should be reported to the client. Log price is almost symmetric (skewness 0.001; Figure 1b).

Table 2 Numerical summaries for all listings, luxury (Audi, BMW, Mercedes) and non-luxury listings

Variable	Group	n	Mean	Median	Mode ^c	SD	Min	Max
Car_Price (INR)	All	1,005,000	1,015,107	673,522	275,000	1,147,780	8,043	51,993,222
	Luxury	232,172	1,012,738	672,614	325,000	1,130,612	11,017	41,975,979
	Non-luxury	772,828	1,015,818	673,731	275,000	1,152,888	8,043	51,993,222
Registration_Age (years)	All	1,005,000	13.0	13.0	6.0	7.2	1.0	25.0
	Luxury	232,172	13.0	13.0	18.0	7.2	1.0	25.0
	Non-luxury	772,828	13.0	13.0	19.0	7.2	1.0	25.0
Kms_Driven (km) ^a	All	998,866	196,142	152,610	95,000	150,624	3,216	999,999
	Luxury	230,726	196,322	153,009	95,000	150,612	3,216	999,919
	Non-luxury	768,140	196,089	152,494	85,000	150,628	3,546	999,999
Engine_CC (cc) ^b	All	1,004,936	1,506	1,494	875	408	304	2,720
	Luxury	232,158	1,507	1,495	875	409	304	2,664
	Non-luxury	772,778	1,506	1,494	875	408	305	2,720
Horsepower (hp) ^b	All	1,004,957	120.5	119.8	62.5	38.7	4.6	235.2
	Luxury	232,159	120.5	119.8	57.5	38.8	7.6	234.0
	Non-luxury	772,798	120.5	119.8	62.5	38.7	4.6	235.2
Mileage_kmpl (km/l) ^b	All	1,004,960	18.00	18.00	11.25	4.18	5.60	30.32
	Luxury	232,164	18.00	17.99	11.25	4.17	5.70	27.11
	Non-luxury	772,796	18.01	18.00	11.25	4.18	5.60	30.32
Accidents (count)	All	1,005,000	0.30	0.00	0.00	0.55	0.00	5.00
	Luxury	232,172	0.30	0.00	0.00	0.55	0.00	5.00
	Non-luxury	772,828	0.30	0.00	0.00	0.55	0.00	5.00

Source: calculated from Stable Space (2026). ^a Kms_Driven $\leq 1,000,000$. ^b Zeros and values beyond mean ± 3 SD excluded; Min and Max are isolated values (Table 1). ^c Exact mode for Registration_Age and Accidents; the continuous variables have no repeated value, so the midpoint of the most frequent class is shown (price 50,000; km 10,000; cc 50; hp 5; km/l 0.5). SD: sample standard deviation.

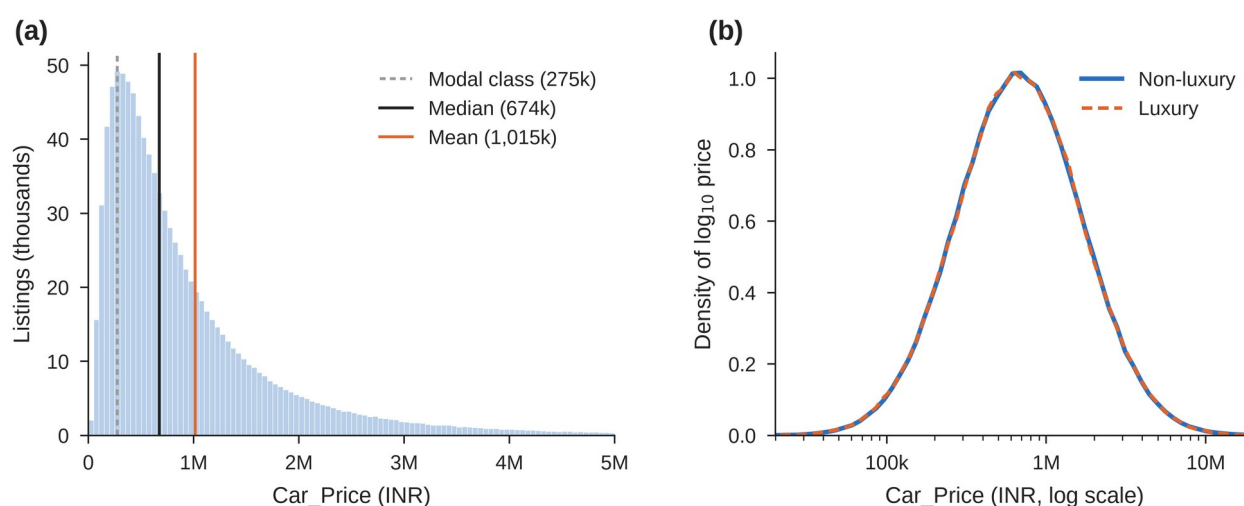


Figure 1 Distribution of Car_Price ($n = 1,005,000$). (a) Raw scale in INR 50,000 classes; the 1.3% of listings above INR 5 million are not shown. (b) Log scale by segment. M = million, k = thousand. Source: calculated from Stable Space (2026).

2.4 Price, vehicle age and usage

Price was plotted for 5,000 randomly chosen listings, the largest sample the guide suggests (UNSW Business School 2026, p. 7), beside medians for all listings (Figure 2). Registration_Age shows no clear association with price ($r = 0.0001$). Its median stays between INR 667,024 and 681,532 across all 25 ages (Figure 2b).

Usage has a weak negative association with price. The correlation is -0.051 , and the median falls from INR 756,694 in the lowest usage decile to INR 603,779 in the highest, 20% lower. This compares different cars, not one car over time. Prices within each decile still vary far more than this (Figure 2d). The median surface in Figure 3a slopes along usage and stays flat along age. Figure 3b shows that chance alone moves r in a 5,000-row sample by about ± 0.028 (95% of samples).

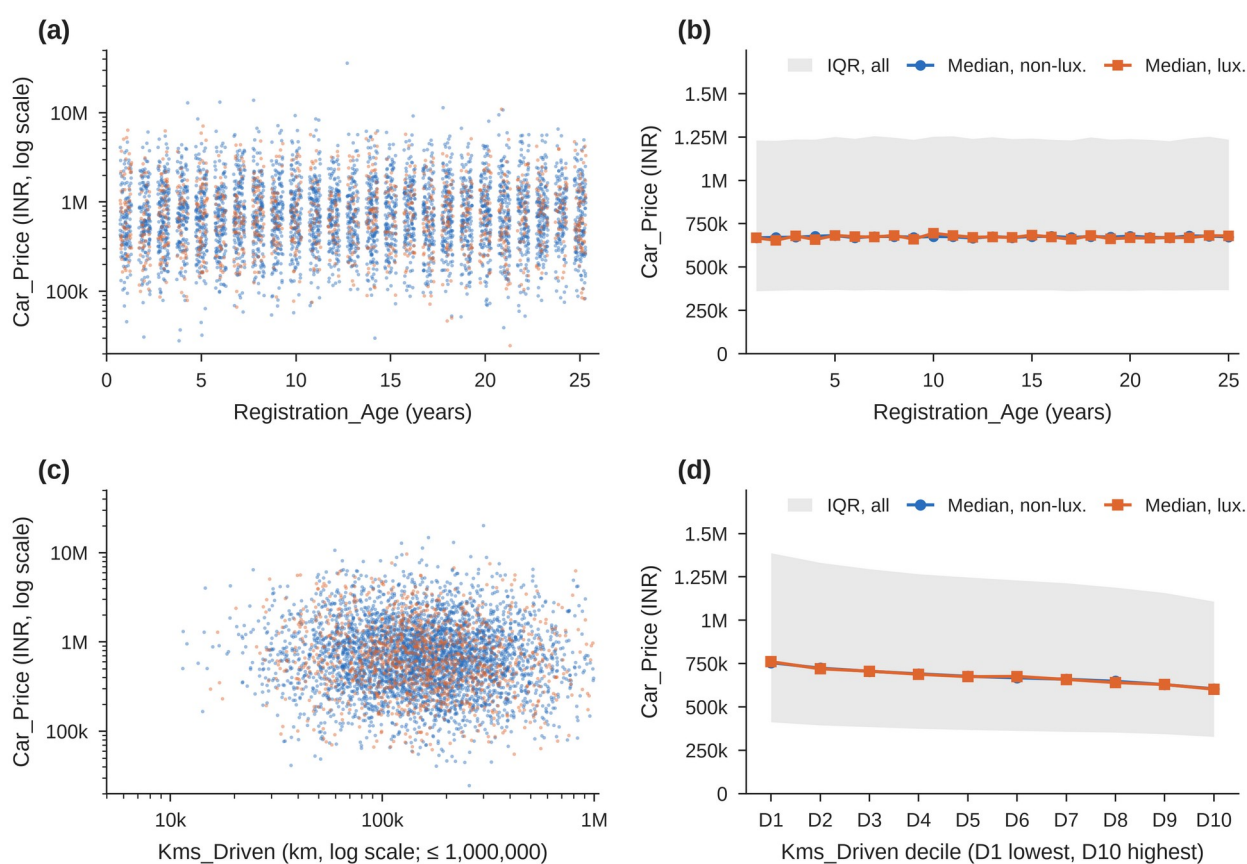
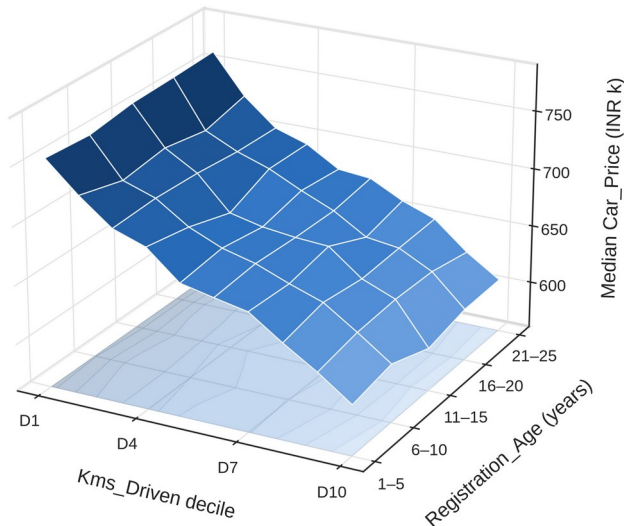


Figure 2 Car_Price against Registration_Age (a, b) and Kms_Driven (c, d). (a) and (c) Random samples of 5,000 listings (orange = luxury, blue = non-luxury); ages in (a) are spread by up to 0.3 years so that points do not overlap. (b) and (d) Medians by segment with the interquartile range (IQR) for all listings; (c) and (d) use listings with Kms_Driven $\leq 1,000,000$. Source: calculated from Stable Space (2026).

(a)



(b)

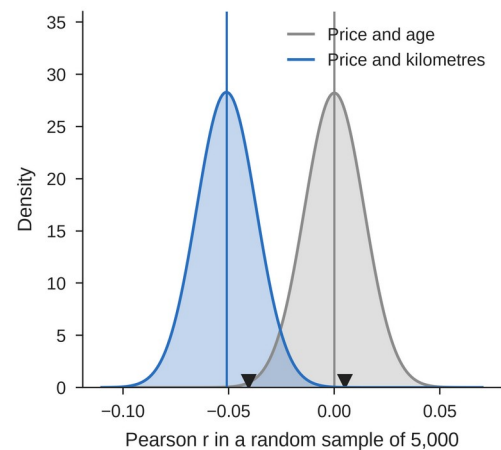


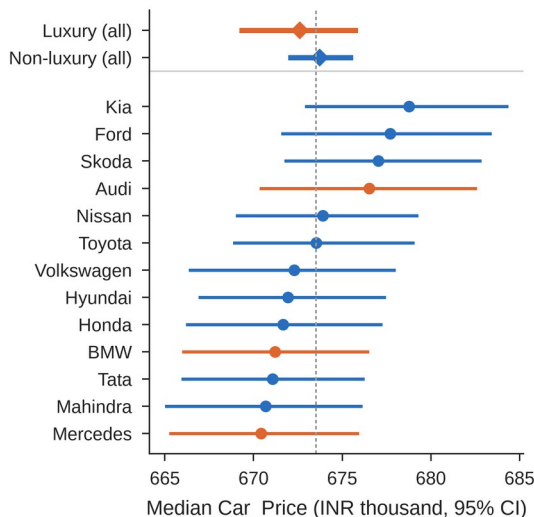
Figure 3 (a) Median Car_Price by five-year age band and usage decile, about 20,000 listings per cell; the vertical axis starts at INR 560,000. (b) Expected spread of the correlation in a random sample of 5,000 rows (normal approximation); vertical lines mark the full-data values and triangles the samples used in Figure 2. Source: calculated from Stable Space (2026).

2.5 Brand segment and other variables

Luxury listings carry no price premium. Their mean price is 0.30% lower than for non-luxury cars and their median 0.17% lower, trivial next to a price SD of INR 1.15 million. The 13 brand medians lie within 1.24% of each other with overlapping intervals (Figure 4a).

Among the other numeric fields, only Horsepower (+0.033) and Engine_CC (+0.026) exceed ± 0.005 in any segment (Figure 4b; Table A2). Other category medians differ by at most 1.53%, and the 3.2% spread across the 39 models is about what chance gives (Figure 5; Table A3). The conclusions hold without Unknown rows, extreme prices or possible duplicates (Table A4).

(a)



(b)

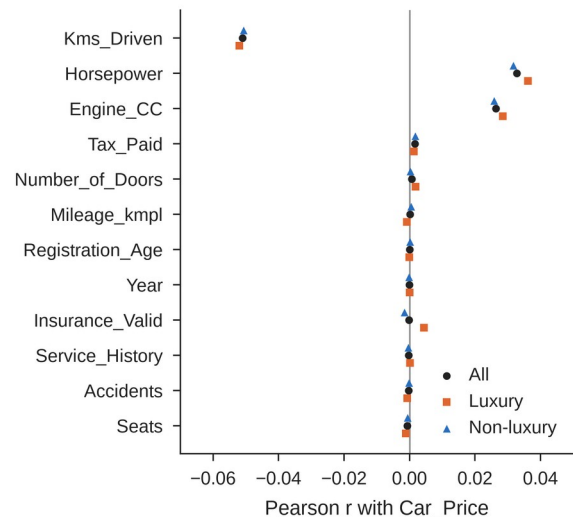


Figure 4 (a) Median Car_Price by brand and segment (orange = luxury, blue = non-luxury) with 95% intervals (ranks $n/2 \pm 1.96\sqrt{n/2}$ of the sorted prices); the axis does not start at zero and the dashed line is the overall median. The 95% confidence interval for the luxury minus non-luxury difference in mean price, as a percentage of the non-luxury mean, is -0.82% to +0.22%. (b) Pearson correlation of each numeric field with Car_Price. Source: calculated from Stable Space (2026).

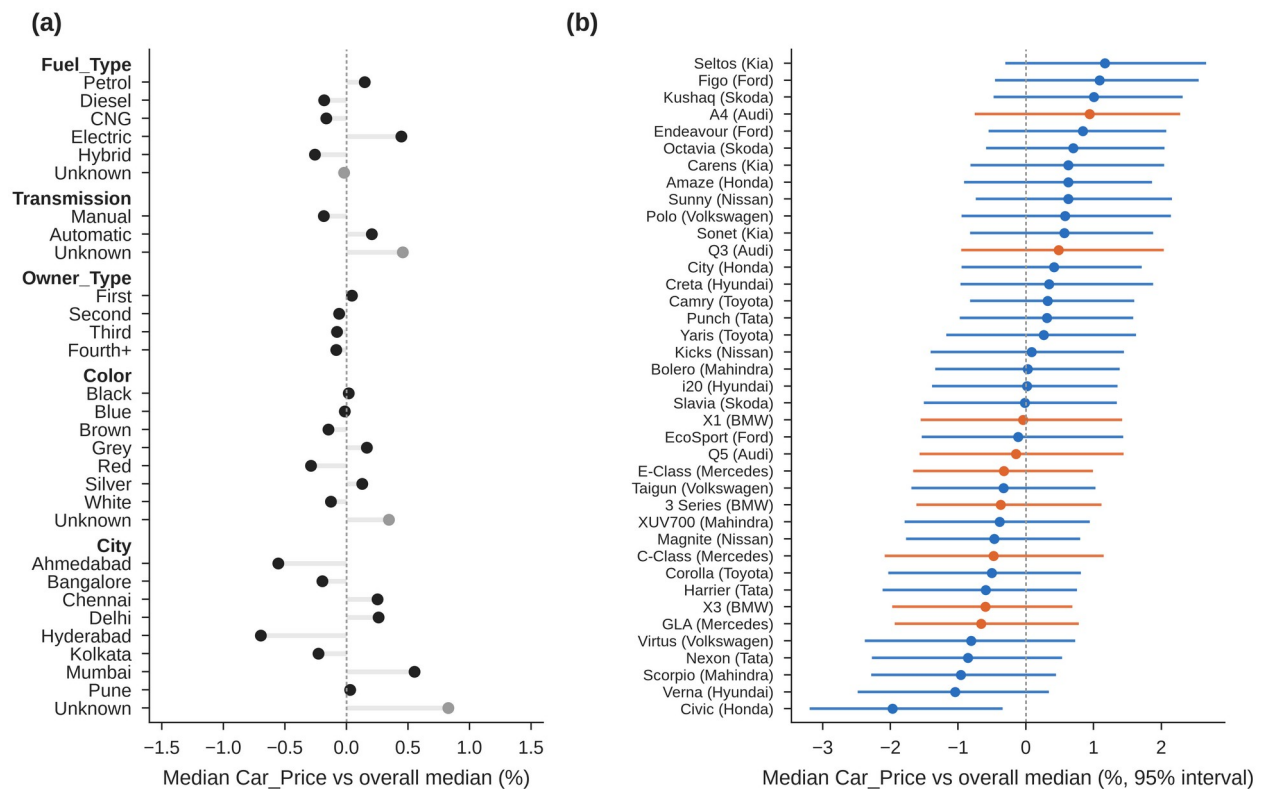


Figure 5 (a) Median Car_Price of each category relative to the overall median (grey = Unknown). (b) Median Car_Price of each model with its 95% interval (ranks $n/2 \pm 1.96\sqrt{n/2}$ of the sorted prices; orange = luxury brand, blue = non-luxury); 38 of the 39 intervals include the overall median. Source: calculated from Stable Space (2026).

3 Conclusion and Development Plan

3.1 Findings

Price should be described by its median and modelled on a log scale. Usage has a weak negative association with price, Horsepower and Engine_CC weaker positive ones, and age, brand segment, model and the other fields none of practical size. A luxury premium in cash offers is therefore not supported. Table 4 sets out recommendations for the client, with their evidence and risks; Table A5 gives the price bands.

Table 4 Recommendations for the client

Recommendation	Evidence	Constraints and risks	Implication
Quote each offer as a median-based price with a range, not a single figure	Skewness 4.9; typical error 55% for the kilometre line (Sections 2.3 and 3.2)	Ranges may look less precise to sellers	Clearer offers; a wide range signals a car that needs an appraiser
Use kilometres only as a small adjustment	$r = -0.051$; about 0.8% lower price per 10% more km; wide spread within each decile (Figure 2d)	The signal is weak and may differ in real sales	Kilometres should not drive an offer on their own
Apply no luxury premium and no separate segment pricing	Luxury mean -0.30% (95% CI -0.82% to $+0.22\%$); brand medians within 1.24% (Figure 4a)	The data are simulated; a real market may show a premium	Re-test on the client's own sales before segment pricing or targeted marketing
Screen listings priced outside the 5th to 95th percentile of their usage decile for an appraiser	Band limits fall with usage (P95 INR 3,339,005 in D1, 2,638,995 in D10); by design 10.0% of listings fall outside (Table A5)	Flags unusual prices, not proven mispricing; bands need refreshing as prices move; listings above 1,000,000 km need their own check	Appraiser time goes to the 10% most unusual prices; other offers keep the standard process until the hold-out test (last row)

Recommendation	Evidence	Constraints and risks	Implication
Tighten data entry: fixed lists for brand and fuel, required fields, no zero specifications, checks for repeats and for more than 1,000,000 km	Table 1: 10,040 brand and 20,123 fuel labels fixed; 283,648 rows with Unknown; about 4,750 excess repeat pairs	Form changes take time; stricter checks may slow listing	Cleaner data and fewer Unknown values for later models
Beyond this project, collect condition, trim and final sale price	The supplied fields explain little of price (Section 3.2)	The project model must use only the supplied variables (Franch 2026, slide 3); collection has a cost	Future models can target sale prices, not asking prices
Automate offers only after a hold-out test and a trial on completed sales	In-sample error 55.2% against 55.3% for the overall median	Delays automation	Avoids systematic over- or under-offers

3.2 Limitations and a better approach

The data are simulated for assessment (UNSW Business School 2026, p. 4), and the macro's code regenerates Mileage_kmpl, Engine_CC, Horsepower, Kms_Driven and Car_Price without reference to brand or age. This probably explains the missing associations. Each summary also looks at one variable at a time. A straight line of log price on log kilometres (Excel SLOPE and INTERCEPT), fitted and checked on the same listings, misses the listed price by a median of 55.2%, barely better than quoting the overall median (55.3%). On that line, 10% more kilometres means a price about 0.8% lower. A better approach would fit a multiple regression on log price, check it on a hold-out sample and quote a price range. Beyond this project, whose model must use only the supplied variables (Franch 2026, slide 3), the client could collect condition, trim and final sale price (Table 4).

3.3 Development plan

Milestone 2 will test the hypotheses in Table 3. The final report will then fit and validate the regression, allowing for the strong Engine_CC–Horsepower correlation ($r = 0.81$).

Table 3 Development plan

Question	Stage	Method	Output
Is vehicle age associated with price?	Milestone 2	H0: the correlation of age and log price is zero; H1: it is not. Compare age bands	Test result and 95% confidence interval
How strongly is usage associated with price?	Milestone 2	H0: the slope of log price on log kilometres is zero; H1: it is not. Compare the two segments	Estimated slope with confidence interval
Is brand segment associated with price?	Milestone 2	H0: mean log price is equal for the two segments; H1: it differs. Two-sample t-test	Test result and 95% confidence interval
Can the fields predict price?	Final report	Multiple regression on log price; fit on 80% of rows, check on the other 20%	Typical error against the median benchmark; price ranges
Do the choices in Table 1 matter?	Both	Repeat key results without Unknown rows, extreme values or possible duplicates	Conclusions kept only if unchanged

References

- Franch, F 2026, 'Used cars: course and Milestone 1 project', lecture slides, COMM5000 Data Literacy, University of New South Wales, Sydney, viewed 26 September 2026, <<https://moodle.telt.unsw.edu.au>>.
- Stable Space 2026, *Used car price prediction dataset*, data set, Kaggle, <<https://www.kaggle.com/datasets/sharmajicoder/used-car-price-prediction-dataset>> (page since removed); randomised copy COMM5000-Used_Car_Price_Prediction.xlsm, UNSW Moodle, viewed 26 September 2026, <<https://moodle.telt.unsw.edu.au>>.
- UNSW Business School 2026, *Assessment guide COMM5000 Data Literacy: case study project, Milestone 1 information*, University of New South Wales, Sydney, viewed 26 September 2026, <<https://moodle.telt.unsw.edu.au>>.

Appendix

Table A1 The 20 fields in the dataset, with the share of listings in each category

Field	Type	Unit or values	Note
Brand	Text	13 brands after trimming	Luxury = Audi, BMW, Mercedes; about 7.7% each
Model	Text	39 models, each with one brand	About 2.6% each
Year	Numeric	2000 to 2024	Year + Registration_Age = 2025
Mileage_kmpl	Numeric	km per litre	Fuel efficiency, not distance
Engine_CC	Numeric	cc	
Horsepower	Numeric	hp	
Fuel_Type	Text	CNG, Diesel, Electric, Hybrid, Petrol, Unknown	Petrol 41.2%, Diesel 27.7%, CNG 9.0%, Electric 7.5%, Hybrid 6.6%, Unknown 7.8%
Transmission	Text	Automatic, Manual, Unknown	Manual 59.8%, Automatic 32.2%, Unknown 8.0%
Owner_Type	Text	First, Second, Third, Fourth+	First 59.9%, Second 25.0%, Third 10.0%, Fourth+ 5.0%
Color	Text	7 colours, Unknown	13.1% to 13.2% each; Unknown 8.0%
City	Text	8 cities, Unknown	11.4% to 11.5% each; Unknown 8.0%
Kms_Driven	Numeric	km	Usage; cap 1,000,000 for analysis
Insurance_Valid	Binary	1 = valid	84.9% valid
Service_History	Binary	1 = records available	75.1% available
Accidents	Numeric	count, 0 to 5	0: 74.1%, 1: 22.2%, 2 or more: 3.7%
Tax_Paid	Binary	1 = paid	90.0% paid
Number_of_Doors	Numeric	2, 4 or 5	4: 60.1%, 5: 34.9%, 2: 5.0%
Seats	Numeric	2, 4, 5 or 7	5: 75.0%, 7: 15.0%, 4: 8.0%, 2: 2.0%
Registration_Age	Numeric	years, 1 to 25	About 40,000 cars at each age
Car_Price	Numeric	INR	Target variable

Source: calculated from Stable Space (2026). Shares are of all 1,005,000 listings; no share differs by more than 0.2 percentage points between the luxury and non-luxury segments.

Table A2 Pearson correlation between Car_Price and each numeric field

Field	All	Luxury	Non-luxury
Kms_Driven ($\leq 1,000,000$ km)	-0.0510	-0.0520	-0.0507
Horsepower	+0.0328	+0.0362	+0.0318
Engine_CC	+0.0264	+0.0285	+0.0258
Tax_Paid	+0.0017	+0.0013	+0.0018
Number_of_Doors	+0.0007	+0.0018	+0.0003
Mileage_kmpl	+0.0002	-0.0008	+0.0005
Registration_Age	+0.0001	-0.0001	+0.0001
Year	-0.0001	+0.0001	-0.0001
Insurance_Valid	-0.0002	+0.0044	-0.0016
Service_History	-0.0003	+0.0002	-0.0004
Accidents	-0.0003	-0.0007	-0.0002
Seats	-0.0007	-0.0011	-0.0006

Source: calculated from Stable Space (2026). Engine_CC and Horsepower are strongly correlated with each other ($r = 0.81$).

Table A3 Median Car_Price across categories (INR)

Field	Categories	Rows per category	Lowest median	Highest median	Spread
Fuel_Type	6	66,793 to 414,245	Hybrid (671,803)	Electric (676,520)	0.70%
Transmission	3	80,385 to 600,630	Manual (672,277)	Unknown (676,611)	0.64%
Owner_Type	4	50,267 to 602,365	Fourth+ (672,954)	First (673,811)	0.13%
Color	8	80,419 to 132,535	Red (671,579)	Unknown (675,846)	0.64%

Field	Categories	Rows per category	Lowest median	Highest median	Spread
City	9	80,430 to 115,999	Hyderabad (668,837)	Unknown (679,097)	1.53%
Model	39	25,524 to 26,046	Civic (660,261)	Seltos (681,372)	3.20%

Source: calculated from Stable Space (2026). Spread = highest median / lowest median – 1. For 38 of the 39 models, the 95% interval of the median contains the overall median.

Table A4 Sensitivity of the main results to the cleaning choices

Scenario	Rows	Median price (INR)	r (age)	r (km)	Luxury median gap
All rows	1,005,000	673,522	+0.00007	–0.0510	–0.17%
Without rows containing Unknown	721,352	672,595	+0.00057	–0.0516	+0.14%
Without the 3 extreme prices	1,004,997	673,521	–0.00002	–0.0513	–0.17%
First row of each 15-field match only	975,360	673,436			–0.19%
Kms cap 500,000	947,751			–0.0487	
Kms cap 750,000	987,792			–0.0503	
Kms cap 1,500,000	1,003,790			–0.0507	
No kms cap	1,005,000			–0.0493	

Source: calculated from Stable Space (2026). The duplicate scenario uses a sorted copy holding only price and segment.

Table A5 Car_Price bands by Kms_Driven decile (INR; listings with Kms_Driven ≤ 1,000,000)

Decile	Kms_Driven (km)	n	P5	Q1	Median	Q3	P95
D1	3,216 to 58,680	99,887	169,069	412,498	756,694	1,387,144	3,339,005
D2	58,680 to 81,556	99,887	163,796	394,299	722,737	1,331,816	3,189,281
D3	81,556 to 103,446	99,886	159,446	384,591	705,564	1,295,140	3,104,325
D4	103,446 to 126,516	99,887	156,335	374,729	690,205	1,265,987	3,045,450
D5	126,516 to 152,610	99,886	154,404	367,915	675,761	1,247,316	3,025,470
D6	152,610 to 184,108	99,887	151,476	362,523	669,288	1,230,314	2,933,993
D7	184,108 to 224,840	99,886	147,982	356,983	659,175	1,213,957	2,916,713
D8	224,840 to 284,119	99,887	146,505	352,845	646,141	1,188,585	2,846,771
D9	284,119 to 390,183	99,886	143,444	343,595	628,605	1,157,695	2,796,179
D10	390,187 to 999,999	99,887	135,828	327,841	603,779	1,108,006	2,638,995

Source: calculated from Stable Space (2026). Listings below P5 or above P95 of their own decile make up 10.0% of the 998,866 listings in this table (screening rule in Table 4).